

Track BC v0.3 — Substrate-Coulomb 1/r derivation framework via Bergman kernel 3D-projection

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Track BC v0.3 — Substrate-Coulomb 1/r via Bergman 3D-projection

1. Cal #29 STANDING question-shape audit (applied at design)

Question: “Does the substrate’s Coulomb potential $f(r) \sim 1/r$ at long range derive forward from Bergman kernel 3D-projection of D_{IV}^5 , independent of the hydrogen 1s observable target?”

Audit: - Structurally determined? YES — Bergman kernel 3D-projection is standard mathematical procedure (substrate’s information-theoretic content projected onto 3D-spatial slice) - Back-fittable? MODERATE RISK — we KNOW Coulomb is $1/r$; tempting to confirm rather than derive - Pre-suppositions? T2442 RIGOROUSLY CLOSED Bergman kernel + standard 3D Green’s function theory

Cal #29 pass with caveat: derivation chain (Bergman kernel $\rightarrow$ 3D projection $\rightarrow$ Green’s function $\rightarrow$ Coulomb $1/r$) must be done with substrate-mechanism content traceable at each step. Honest negative outcome possible: if 3D-projection of Bergman kernel doesn’t give $1/r$ cleanly, substrate-Coulomb may have more complex form than standard QED predicts.

2. The framework — Bergman kernel as Green’s function source

2.1 Bergman reproducing kernel on D_{IV}^5

Per T2442 RIGOROUSLY CLOSED:

$$K(z, \bar{w}) = c_{FK} \cdot (1 - \langle z, \bar{w} \rangle)^{(-g/\text{rank})} = c_{FK} \cdot (1 - \langle z, \bar{w} \rangle)^{(-7/2)}$$

with $c_{FK} \cdot \pi^{(9/2)} = 225$ EXACT.

D_{IV}^5 is 5-complex-dim (10-real-dim) bounded symmetric domain. The Bergman

kernel reproduces holomorphic functions: for any $f \in H^2(D_{IV}^5)$, $f(z) = \int_{D_{IV}^5} K(z, \bar{w}) \cdot f(w) d\nu(w)$

with $d\nu$ Hua-Loos measure.

2.2 3D-projection of D_{IV}^5

The substrate is 10-real-dim D_{IV}^5 ; observable physics is 3+1-D Minkowski (or 3-D spatial slice). The 3D-projection map $\pi: D_{IV}^5 \rightarrow \mathbb{R}^3$ takes substrate states to 3D-spatial observables.

Substrate-mechanism: per SWPP RATIFIED Casey-named + Paper #122 + T2467+T2468 Rigidity Principle, the substrate produces 3+1-D spacetime as PROJECTION of its 10-real-dim D_{IV}^5 structure. The Bergman kernel 3D-projection gives the substrate-induced 3D Green's function.

Standard projection structure (substrate-natural candidate): - D_{IV}^5 has 5-complex-dim Hua coordinates $z = (z_1, z_2, z_3, z_4, z_5)$ - 3D-projection: real parts of z_1, z_2, z_3 (or some Hua-aligned subset) $\rightarrow$ 3 spatial coordinates - Imaginary parts + remaining real parts $\rightarrow$ substrate-internal/transverse content

2.3 The Green's function

The substrate-induced 3D Green's function $G(x, y)$ for $x, y \in \mathbb{R}^3$ is the 3D-projection of the Bergman kernel:

$$G(x, y) = \pi^* K(z(x), \bar{w}(y)) \text{ [3D-projection notation]} = c_{FK} \cdot \pi^* [(1 - \langle z(x), \bar{w}(y) \rangle)^{-7/2}] \text{ [explicit form]}$$

For $r = |x - y|$ small relative to substrate cutoff, this 3D-projected kernel should reduce to standard 3D Green's function $\sim 1/r$ (Coulomb-like) by general principles.

3. The substrate-Coulomb derivation framework

3.1 Far-field limit

For $r \gg L_K \approx 10^{(-112)} \text{ m}$ (substrate-tick length scale), the 3D-projection should reduce to standard 3D scalar Green's function:

$$G(r) = 1/(4\pi \cdot r)$$

at far-field. This is the standard Coulomb potential structure: $f(r) = q_p \cdot \alpha / r$ (for proton charge $q_p = +1$ and electron test charge).

Substantive substrate-mechanism question: does Bergman kernel 3D-projection of D_{IV}^5 reproduce $1/r$ at far-field by mathematical necessity, or does it produce a different functional form?

Hua-Loos structural argument: $\pi^* K(z, \bar{w})$ for D_{IV}^5 at far-field is asymptotically: $K(z, \bar{w}) \sim (1 - \langle z, \bar{w} \rangle)^{-7/2} \sim |1 - \langle z, \bar{w} \rangle|^{-7/2}$

For 3D-projected coordinates with $|x - y| = r$ large: $\langle z(x), \bar{w}(y) \rangle \rightarrow c_\infty(r)$ which is r -dependent. Standard projection theorems (multi-week explicit work): - $\langle z(x), \bar{w}(y) \rangle \rightarrow$ some function of r - Power law $(1 - f(r))^{(-7/2)}$ at $r \rightarrow \infty \rightarrow$ asymptotic r -dependence

The substantive question: does the asymptotic r -dependence give exactly $1/r$, or something else?

Cal #133 tautology audit: this is the kind of question where we KNOW the target ($1/r$ Coulomb) and the framework should produce it. Cal #29 design audit: forward-derivation discipline — derive asymptotic form from Bergman 3D-projection FIRST, THEN compare to $1/r$.

3.2 Near-field substrate-tick modifications

For $r \sim L_K$ (Planck-scale or substrate-tick scale), substrate-mechanism modifications enter: - Discreteness at substrate-tick scale: Green's function gets cyclotomic K59 corrections - 7-step cyclotomic chain modifies short-range behavior - BST-primary structure ($137 = N_{\max}$) sets natural UV cutoff

At atomic scale ($r \sim a_0 \approx 10^{(-11)} \text{ m}$), $r \gg L_K$, so substrate-tick corrections are negligibly small (suppressed by $(L_K/r)^?$ — some substrate-mechanism power).

v0.4+ explicit computation: derive substrate-tick correction explicit form; verify negligibility at atomic scale.

3.3 α -quantization of charge coupling

Per T2447 RIGOROUSLY CLOSED + T2476 STRUCTURALLY VERIFIED: substrate's electromagnetic coupling is $\alpha = 1/N_{\max} = 1/137$. Per SWPP + Reed-Solomon K59 mechanism: the coupling is α -quantized at substrate-tick scale.

Substrate-mechanism for Coulomb potential: $f(r) = q_p \cdot \alpha / r$ (long-range, after Bergman 3D-projection) = $+1 \cdot (1/137) \cdot (1/r) = 1/(137 r)$ (proton at origin, test charge +1, in substrate units)

The α factor emerges from substrate's $N_{\max} = 1/\alpha$ structure (T2447). The $1/r$ emerges from Bergman 3D-projection (Section 3.1).

3.4 The Shilov boundary condition for hydrogen 1s

Combining: the proton's substrate K-type $V_{(3/2, 1/2)}$ at spatial position R_p imposes Shilov boundary condition:

$$\varphi_{\{\text{proton-BC}\}}(\bar{w}; r) = \exp(i \cdot q_{\text{electron}} \cdot \alpha \cdot (1/r) \cdot \text{scale_factor}) = \exp(-i \cdot \alpha/r \cdot \text{scale_factor}) = \exp(-i \cdot (1/137 \cdot r) \cdot \text{scale_factor})$$

where scale_factor is a substrate-natural length unit (atomic-scale anchor; v0.4 derives explicitly).

Bergman integral (v0.4 multi-week explicit): $\psi_{\{1s\}}(z) = \int \{S^4 \times S^1\} c_{FK} \cdot (1 - \langle z, \bar{w} \rangle)^{-7/2} \cdot V(1/2, 1/2)(\bar{w}) \cdot \exp(-i \alpha / r(\bar{w})) d\mu(\bar{w})$

4. The α^2 binding factor — structural emergence (v0.2 carried forward)

Per v0.2 Section 5.2: the hydrogen 1s ground state energy $E_{1s} = -\alpha^2/2 \cdot m_e c^2$ emerges with TWO factors of α :

1. One α from substrate-Coulomb phase (Section 3.3): α factor in $f(r) = \alpha/r$
2. One α from Bergman 7/2 weighting on Shilov boundary: integration over Shilov boundary picks up α -quantized factor via $\text{Pin}(2) Z_2$ phase content

Product: $\alpha \cdot \alpha = \alpha^2 \rightarrow E_{1s}$ scaling matches observed $-\alpha^2/2 \cdot m_e c^2$.

v0.3 strengthening: the “first α ” is now substrate-mechanism-derived (T2447 $N_{\max} = 1/\alpha$) + Bergman 3D-projection. The “second α ” via Bergman 7/2 weighting requires multi-week derivation chain at the Shilov boundary integration step.

5. Multi-week explicit derivation path (v0.4+)

5.1 Bergman 3D-projection asymptotic form (v0.4)

Explicit computation: - 3D-projection $\pi: D_{IV}^5 \rightarrow \mathbb{R}^3$ choice (Hua coordinate selection) - Bergman kernel asymptotic form $(1 - \langle z, \bar{w} \rangle)^{-7/2}$ at far-field - Verify: does it reduce to $1/(4\pi r)$ or some other functional form? - Multi-week analytical computation.

5.2 Substrate-Coulomb explicit form (v0.5)

If v0.4 confirms $1/r$ at far-field: - Explicit form $f(r) = q \cdot \alpha \cdot h(r/L_K)$ with $h(\infty) = 1/r$ and $h(0) =$ substrate-tick limit - BST-primary structure of $h(r/L_K)$ function - Multi-week.

5.3 Bergman integral evaluation for hydrogen 1s (v0.6)

With substrate-Coulomb $f(r)$ explicit: - Evaluate Bergman integral - Express in standard special functions (hypergeometric likely) - Compare to $\psi_{\{1s\}}(r) = (1/\sqrt{(\pi a_0^3)}) e^{-(r/a_0)}$ - Multi-week.

5.4 α^2 -binding numerical verification (v0.7)

- Compute $\langle \psi_{\{1s\}} | \hat{H}_{\text{sub}} | \psi_{\{1s\}} \rangle$
- Verify $E_{1s} = -\alpha^2/2 \cdot m_e c^2$ emergence
- Multi-week.

5.5 Higher-n extension + multi-electron (v0.8+)

- 2s, 2p, $n \geq 2$ hydrogen states via reaction-table edges
- Helium, lithium, molecular hydrogen
- Multi-month per extension.

6. Cal Thread 4 typing for v0.3 substrate-Coulomb

Per Cal #122 Type A geometric / Type B algebraic / Type C level-crossing:

Substrate-Coulomb via Bergman 3D-projection: - Bergman kernel is Type A geometric (substrate geometry on D_{IV}^5) - 3D-projection is Type A geometric (projection between substrate and observable geometries) - α factor via T2447 is Type C level-crossing (substrate $N_{\max} = 1/\alpha$ connects substrate-geometric structure to observable electromagnetic coupling) - Combined substrate-Coulomb $1/r$ is Type C level-crossing (substrate geometry $\times$ observable electromagnetic coupling)

My prior: Type C level-crossing. Substrate-Coulomb is the level-translation from substrate D_{IV}^5 to observable 3D Coulomb law.

7. Connection to Track DC rank-2 signature

The substrate-Coulomb derivation framework (Track BC v0.3) and the Bell $1/8$ rank-2 signature (Track DC v0.3) share substrate-mechanism components:

- Both use T2442 Bergman kernel at substrate algebra level (Bergman normalization 225)
- Both use T2447 $\alpha = 1/N_{\max}$ at substrate coupling level (electromagnetic + Bell-CHSH)
- Both involve 3D-projection structure (Bell test in 3D space; Coulomb potential in 3D space)
- Both reduce to substrate-mechanism rather than back-fitted numerical match

The Tracks are convergent at substrate-mechanism level. If both v0.3+ multi-week derivations close, the BST framework gains two substantial mechanism derivations: Bell $1/8$ from rank-2 signature, Coulomb $1/r$ from Bergman 3D-projection.

8. Honest scope (Cal #27 STANDING + Cal #29 STANDING + Cal #133)

What's RATIFIED / SVC: - Bergman kernel $K(z, \bar{w}) = c_{FK} \cdot (1 - \langle z, \bar{w} \rangle)^{(-7/2)}$ (T2442 RIGOROUSLY CLOSED) - $\alpha = 1/N_{\max} = 1/137$ (T2447 RIGOROUSLY CLOSED) - Standard 3D Green's function $1/(4\pi r)$ (mathematical fact)

What's FRAMEWORK in v0.3: - Bergman 3D-projection structure (Section 2.2) - Substrate-Coulomb potential framework (Section 3.3) - α^2 binding factor structural emergence (Section 4)

What's INTERPRETIVE in v0.3 (Cal #133 + Cal #29 risk-flag preserved): - "Substrate-Coulomb $1/r$ emerges from Bergman 3D-projection" — multi-week explicit verification needed (Section 5.1) - 3D-projection map $\pi: D_{IV}^5 \rightarrow \mathbb{R}^3$ explicit form — multi-week - Substrate-tick correction explicit form — multi-week

What's NOT in v0.3 (multi-week+): - Explicit Bergman 3D-projection computation (v0.4) - Substrate-Coulomb $f(r)$ verification at $1/r$ (v0.4) - Bergman integral evaluation for hydrogen $1s$ (v0.6) - α^2 -binding numerical verification (v0.7)

Cal #27 STANDING reflexive trigger count: 1 trigger (Section 4 α^2 binding double- α emergence). Flagged FRAMEWORK pending Section 5.1-5.4 explicit derivation.

Cal #29 STANDING pass with caveat: Bergman 3D-projection derivation is structural; forward-derivation discipline applies (derive asymptotic form FIRST, then verify against $1/r$ target). Honest negative outcome possible.

9. Coordination

Cal: Thread 4 cold-read on Section 6 type-check (Type C level-crossing prior) + Section 5.1 explicit derivation framework + Section 4 α^2 binding interpretive content.

Elie: Toy 3539+ candidate — explicit Bergman 3D-projection numerical evaluation. Cal #29 pre-audit required.

Grace: catalog cross-references for substrate-Coulomb + hydrogen-related observables; Phase 2 SPLP audit (launched 11:14 EDT) may encounter hydrogen entries.

Keeper: integration; Vol 15 Ch 9 case study draft includes Track BC v0.3 substrate-Coulomb framework.

— Lyra, Track BC v0.3 substrate-Coulomb $1/r$ via Bergman 3D-projection framework filed Tuesday 2026-05-26 ~11:25 EDT per Casey no-pause + all-tracks authorization. FRAMEWORK grade. Establishes structural framework for substrate-Coulomb potential derivation via Bergman kernel 3D-projection of D_{IV}^5 ; α^2 binding factor structural emergence with one α from T2447 $N_{\max} = 1/\alpha$ + one from Bergman $7/2$ weighting. Multi-week explicit derivation path identified (v0.4 Bergman 3D-projection asymptotic form $\rightarrow$ v0.5 substrate-Coulomb explicit $\rightarrow$ v0.6 Bergman integral $\rightarrow$ v0.7 α^2 -binding numerical $\rightarrow$ v0.8+ extension). Cal Thread 4 type-check: Type C level-crossing prior (substrate geometry $\times$ observable electromagnetic coupling).